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Differential-steering based path tracking control and energy-saving torque distribution strategy of 6WID unmanned ground vehicle

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A B S T R A C T

The unmanned ground vehicle (UGV) is developing towards miniaturization, lightweight and intelligence. Path tracking and energy-saving control strategies are of great significance. In this study, a hierarchical control strategy is proposed to realize the path tracking control of six-wheel independent drive UGV (6WID UGV) based on differential steering, where actuator fault and energy-saving are considered. Firstly, a model free adaptive sliding mode control (MFASMC) method is utilized in the upper controller to estimate the required yaw moment. Then, in the lower controller, the seeker optimization algorithm (SOA) is utilized to achieve the torque distribution, realize the required yaw moment and longitudinal force, and reduce the energy consumption in the process of path tracking. The fault-tolerant control is considered in the torque distribution strategy to ensure the stability of 6WID UGV in case of motor execution fault. Finally, the effectiveness of the proposed scheme is verified by simulation under various operating conditions. The results show that the proposed method has better control performance, and the characteristics of energy-saving are optimized while ensuring the vehicle stability.

1. Introduction

Unmanned ground vehicle (UGV) refers to a robot system that runs on the ground and can perform specific tasks. It is a highly integrated mobile vehicle of mechanization, informatization and intelligence [1]. The UGV can complete a variety of tasks that are difficult for humans in high-risk environment, such as disaster recovery, localization of wounded personal, transportation of materials, sample collection and data acquisition. It requires the ability of all terrain rapid mobility and long mileage [2]. With the development of the driving system of UGV, a six-wheel independent driving unmanned ground vehicle is proposed, which has good prospects in military and civil fields [3]. Independent drive system [4,5] and independent steering mechanism provide a variety of steering modes for 6WID UGV. Based on the hybrid electric system [6–8] and variable configuration [1–3], the 6WID UGV has higher mobility and environmental adaptability. The UGV needs to gradually have the ability of autonomous control for task execution, and its path tracking control is particularly significant [9].

The 6WID UGV has the characteristics of high-frequency start/stop and heavy-duty steering when performing tasks in the field. Differential steering can be used to ensure the stability when the independent steering fails. Although differential steering relies on the speed difference between the wheels on both sides of the vehicle and the tire wear is more obvious than independent steering, the 6WID UGV is driven by in-wheel motors, which make the differential steering control more flexible. At present, in the fields of autonomous vehicles, there have been a large number of research results in the active steering system [10] and the path tracking control method based on independent steering [11,12]. Therefore, in order to comprehensively improve the reliability and adaptability to complex operating environment and emergencies of 6WID UGV, it is very important to investigate the path tracking control on the basis of differential steering mode.

The key issue of differential steering [13] of 6WID UGV is to flexibly distribute the torque of six in-wheel motors to realize different speed of driving wheels. In order to realize yaw control when the active front steering fails completely, Hu et al. proposed an integral sliding mode control method for independent drive

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vehicles [14]. In Ref. [15], the differential drive method of independent drive articulated heavy vehicles based on model predictive control is studied. When the active steering motor completely fails, in order to ensure safety, Hu et al. studied the lane keeping control of four-wheel independent drive autonomous vehicles [16], and the steering control is realized by the differential torque between the front wheels. Torque distribution can not only ensure the stability of the vehicle, but also decrease the energy consumption [17]. Kwon et al. analyzed the effects of torque distribution and transmission ratio on motor efficiency for electric vehicles, a multi-objective optimization method was proposed to give consideration to performance and efficiency [18]. In Ref. [19], the control strategy based on game theory is proposed to coordinate the motor, internal combustion engine and vehicle braking system. In Ref. [20], the real-time optimal torque distribution strategy based on the prediction of driving conditions is studied. Based on the research results mentioned above, the following conclusions can be obtained: (1) existing torque distribution strategy mainly focusses on the stability and energy efficiency of manned vehicles. Nevertheless, facing the influence of a complex environment, the torque distribution strategy of 6WID UGV should not only consider stability and energy optimization, but also meet the requirements of autonomous control. (2) few studies have focused on the path tracking control of UGV using differential steering, and the energy optimization in the process of differential steering has not been considered. Importantly, UGV is also developing towards miniaturization and lightweight. Considering energy-saving optimization in process of path tracking has an important impact on improving the mileage and execution efficiency of 6WID UGV.

Although the 6WID UGV adopts the in-wheel motors and has the advantage of flexibility, it also faces the risk of motor execution fault [21] in the process of differential steering. In order to ensure the stability of the vehicle when the in-wheel motor or steering system fails, a three-step fault-tolerant control strategy is studied in Ref. [22]. Chen et al. studied the active fault-tolerant control strategy based on integral sliding mode control for a class of uncertain nonlinear systems [23]. In Ref. [24], an active fault-tolerant control strategy for independent drive vehicles is proposed, which combines active fault-tolerant control and three-step control to deal with actuator fault. To ensure the stability of four-wheel independent drive electric vehicle, a new fault-tolerant control method based on game theory is proposed in Ref. [25]. In Ref. [26], a discrete-time fault-tolerant control strategy is proposed for independent drive electric ground vehicles with active steering system, considering the adhesion limitation of tires and the impact of potential faults on in-wheel motors and active steering system. Compared with these contributions, the 6WID UGV has more in-wheel motors, and the problem of actuator redundancy is more serious. When the in-wheel motor fault occurs, how to ensure the stability and design the control allocation strategy of 6WID UGV still needs to be further investigated.

Consequently, this paper studies the path tracking control and torque distribution strategy of 6WID UGV based on differential steering, to ensure the stability of the vehicle and optimize the energy-saving characteristics. The main work of this paper is as follows: (1) a model free adaptive sliding mode control (MFASMC) scheme is proposed to estimate the required yaw moment; (2) a torque distribution strategy with fault-tolerant control based on seeker optimization algorithm (SOA) is proposed to reduce the energy consumption in the process of path tracking while ensuring vehicle stability; (3) the feasibility and effectiveness of proposed method are proved by co-simulation results.

The rest paper is organized as follows. In Section 2, the structure of 6WID UGV is presented. The multi axle vehicle model, the differential steering model and the in-wheel motor model are established; In Section 3, a hierarchical control structure is presented and a differential-steering based path tracking control strategy with fault-tolerant control is proposed. the desired longitudinal force and yaw moment are estimated in upper controller, and the lower controller distributes the torque to six in-wheel motors; In Section 4, the path tracking performance and energy-saving characteristics under different working conditions are simulated and analyzed; Finally, Section 5 concludes the paper and outlines areas for future research.

2. System composition and modeling

The 6WID UGV [1–3] uses the hydraulic system to realize variable wheelbase and suspension height to improve the terrain adaptability. When the independent steering system fails, the differential steering can ensure the driving stability and achieve autonomous control. The 6WID UGV adopts the in-wheel motors, which can flexibly control the torque of each wheel independently. Accordingly, this paper studies the path tracking control strategy based on differential steering of 6WID UGV. By optimizing the energy consumption in the path tracking process and considering the fault-tolerant control, it can comprehensively improve the reliability and adaptability to complex operating environment and emergencies. The driving system, braking system, and research content of the 6WID UGV are shown in Fig. 1.

2.1. Vehicle model

The dynamic model of 6WID UGV is the basis for studying its stability and torque distribution strategy. The differential steering model and 2 DOF model are shown in Fig. 2.

As shown in Fig. 2 (a), the motion of 6WID UGV is affected by longitudinal force, lateral force, and yaw moment. According to 2DOF linear model [27], the dynamic equation of 6WID UGV can be obtained as follows.

$$m v_x(\dot{\beta} + r) = F_{Yf} + F_{Ym} + F_{Yr} \tag{1}$$

$$I_z \dot{\gamma} = l_f F_{Yf} - l_m F_{Ym} - l_r F_{Yr} + M_z \tag{2}$$

where m is the mass of the vehicle; v_x represents the longitudinal speed; β is the sideslip angle; γ is the yaw rate; F_{Yf},F_{Ym} and F_{Yr} are the lateral force of each axle; I_z is the inertia; l_f, l_m and l_r respectively represent the positions of the front axle, middle axle and rear axle to the center of gravity, where it is assumed that the middle axle is behind the center of gravity. M_Z is the external yaw moment generated by the longitudinal force, which can be described by the following.

$$M_z = l_b(F_{xr} - F_{xl}) \tag{3}$$

where F_{xr} and F_{xl} respectively represent the total longitudinal forces provided by in-wheel motors on both sides of 6WID UGV; l_b represents half of the track width. The total longitudinal force provided by the drive system and the brake system can be expressed as

$$F_x = m\frac{dv_x}{dt} + F_f \tag{4}$$

where F_x represents the total longitudinal forces; F_f is the total resistance.

The vertical tire-road force is an important factor affecting the steering performance of 6WID UGV, and it is also the key constraint of torque distribution. According to Ref. [3], the vertical tire-road force of each axle under uniform speed condition is

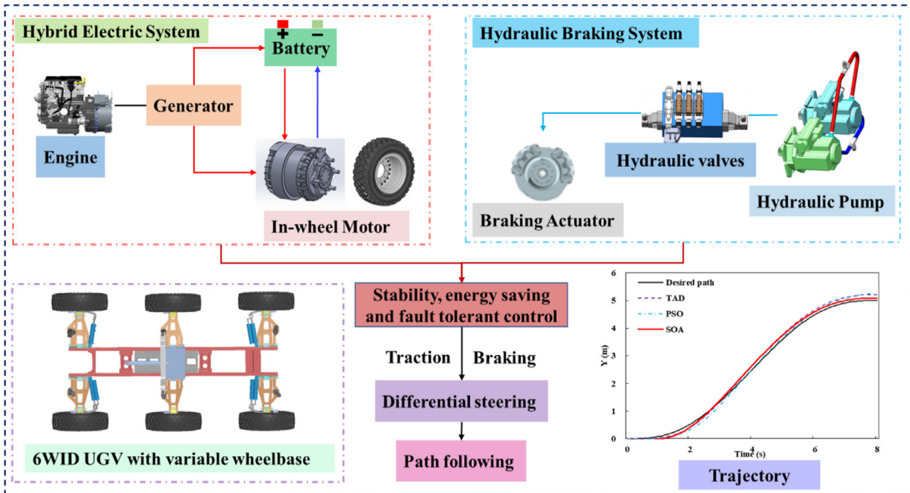


Fig. 1. Structure of 6WID UGV

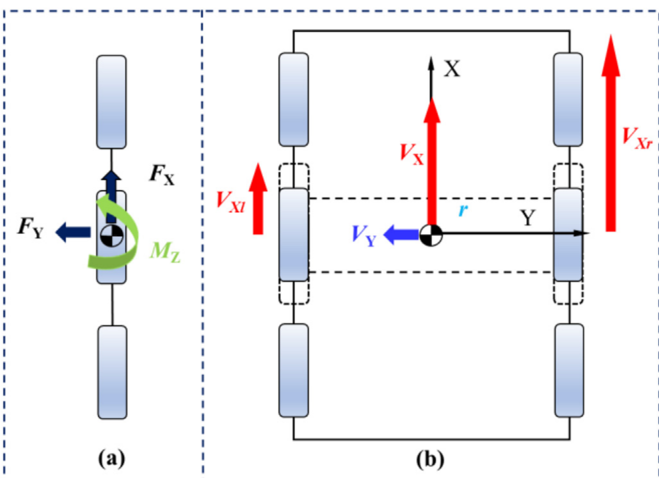


Fig. 2. Vehicle dynamics and differential steering model.

$$F_{zf} = \frac{mg}{4}\frac{l_f l_r + l_m l_f + l_m^2 + l_r^2}{l_f^2 + l_m^2 + l_r^2 + l_m l_f + l_r l_f - l_m l_r} \tag{5}$$

$$F_{zm} = \frac{mg}{4}\frac{l_m l_f - l_m l_r + l_f^2 + l_r^2}{l_f^2 + l_m^2 + l_r^2 + l_m l_f + l_r l_f - l_m l_r} \tag{6}$$

$$F_{zr} = \frac{mg}{4}\frac{l_f l_r - l_m l_r + l_m^2 + l_f^2}{l_f^2 + l_m^2 + l_r^2 + l_m l_f + l_r l_f - l_m l_r} \tag{7}$$

When the vehicle has both longitudinal and lateral accelerations, the load force of each tire can be expressed as

$$F_{zfl} = F_{zf} + \frac{ma_x h}{2}\frac{-2l_f - l_m - l_r}{(l_r - l_m)(2l_r - l_m) + 2\left(l_f + l_m\right)^2} - \frac{ma_y h}{3l_b} \tag{8}$$

$$F_{zfr} = F_{zf} + \frac{ma_x h}{2}\frac{-2l_f - l_m - l_r}{(l_r - l_m)(2l_r - l_m) + 2\left(l_f + l_m\right)^2} + \frac{ma_y h}{3l_b} \tag{9}$$

$$F_{zml} = F_{zm} + \frac{ma_x h}{2}\frac{l_f - l_r + 2l_m}{(l_r - l_m)(2l_r - l_m) + 2\left(l_f + l_m\right)^2} - \frac{ma_y h}{3l_b} \tag{10}$$

$$F_{zmr} = F_{zm} + \frac{ma_x h}{2}\frac{l_f - l_r + 2l_m}{(l_r - l_m)(2l_r - l_m) + 2\left(l_f + l_m\right)^2} + \frac{ma_y h}{3l_b} \tag{11}$$

$$F_{zrl} = F_{zr} + \frac{ma_x h}{2}\frac{l_f + 2l_r - l_m}{(l_r - l_m)(2l_r - l_m) + 2\left(l_f + l_m\right)^2} - \frac{ma_y h}{3l_b} \tag{12}$$

$$F_{zrr} = F_{zr} + \frac{ma_x h}{2}\frac{l_f + 2l_r - l_m}{(l_r - l_m)(2l_r - l_m) + 2\left(l_f + l_m\right)^2} + \frac{ma_y h}{3l_b} \tag{13}$$

where F_{zi} [i = fl,fr,ml,mr,rl,rr] represents the vertical tire-road force of each tire; a_x and a_y denote the longitudinal and lateral accelerations respectively; h is height of center of gravity.

The characteristics of tire are the basis of studying vehicle dynamics, which have a very important impact on vehicle handling stability. Tire model can describe the interaction between tires and road surface, and directly determine the movement of vehicles. Therefore, many tire models have been proposed in the existing research literature, and "Magic Formula (MF)" model is the most widely used. The general form of MF can be expressed as

$$y(x) = D\sin\{C\arctan[Bx - E(Bx - \arctan Bx)]\} \tag{14}$$

where y(x) represents longitudinal force or lateral force; the variable x is the longitudinal slip ratio or lateral sideslip angle; B is the stiffness factor; D is the peak factor; C is the shape factor; E is the curvature factor.

Assuming that the tire works in the linear region [1–3], the dynamic model of 6WID UGV can be expressed as

$$\begin{bmatrix}\dot{\theta}\\ \dot{\psi}\\ \dot{\beta}\\ \dot{\gamma}\end{bmatrix}=\begin{bmatrix}0 & 0 & \dfrac{2\left(K_f+K_m+K_r\right)}{mv_x} & \dfrac{2\left(K_fl_f-K_ml_m-K_rl_r\right)}{mv_x^2}\\ 0 & 0 & 0 & 1\\ 0 & 0 & \dfrac{2\left(K_f+K_m+K_r\right)}{mv_x} & \dfrac{2\left(K_fl_f-K_ml_m-K_rl_r\right)}{mv_x^2}-1\\ 0 & 0 & \dfrac{2\left(K_fl_f-K_ml_m-K_rl_r\right)}{I_z} & \dfrac{2\left(K_fl_f^2+K_ml_{l_m}^2+K_rl_r^2\right)}{I_zv_x}\end{bmatrix}\times\begin{bmatrix}\theta\\ \psi\\ \beta\\ \gamma\end{bmatrix}+\begin{bmatrix}0\\ 0\\ 0\\ \dfrac{1}{I_z}\end{bmatrix}M_z \tag{15}$$

where K_f, K_m and K_r are the cornering stiffness of front, middle and rear wheels respectively; θ and ψ are the heading angle and yaw angle respectively, and $\theta = \psi + \beta$. Although the vehicle model is simplified, it can still be seen that 6WID UGV is a strong nonlinear system. Therefore, this paper adopts the idea of model-free control to design its path tracking and torque distribution strategy. The above model is only used to determine the output of the controllers.

2.2. Path tracking and differential steering model

According to Eq. (15), the state equation of heading angle can be described as

$$\dot{\theta}=\frac{2\left(K_f+K_m+K_r\right)}{mv_x}\beta+\frac{2\left(K_fl_f-K_ml_m-K_rl_r\right)}{mv_x^2}\gamma \tag{16}$$

By constructing the desired heading angle function, the complex path tracking problem is simplified to a simple heading tracking problem [1]. When the heading error converges to zero, the path error also converges to zero and the tracking error can be defined as

$$e=\theta_{des}-\theta \tag{17}$$

Fig. 2 (b) illustrates the principle of differential steering of 6WID UGV. The rotation speed of wheels on both sides is different through torque control to achieve the purpose of steering. The instantaneous velocity of the center of gravity of the 6WID UGV can be expressed as

$$v_x=\frac{1}{2}\left(v_{xl}+v_{xr}\right). \tag{18}$$

where v_{xr} and v_{xl} respectively represent the speed of right and left wheels.

Regardless of the original steering, the steering radius is as follows.

$$R=\frac{l_b\left(v_{xr}+v_{xl}\right)}{2\left(v_{xr}-v_{xl}\right)} \tag{19}$$

where R is the steering radius.

2.3. In-wheel motor and braking model

In-wheel motor is the main executive component of drive, regenerative braking and differential steering. The motor model can be simplified into a first-order system [28]. In-wheel motor and reducer are integrated in the wheel, so the torque of the electric wheel is expressed as

$$T_{ea}=\frac{1}{\tau s+1}T_{ed}, T_{e\min}\le T_{ed}\le T_{e\max} \tag{20}$$

$$F_{xi}=\left(T_{ea}-J\frac{\mathrm{d}\omega_{xi}}{\mathrm{d}t}\right)\Big/ r_i \tag{21}$$

where T_{ea} is the actual output torque of electric wheel; τ is time constant; T_{ed} is the desired torque calculated by allocation algorithm; $T_{e\max}$ and $T_{e\min}$ are the maximum and minimum torque of the electric wheels, respectively; F_{xi} represents the longitudinal force of each in-wheel motor; J is the inertia; ω_{xi} indicates the speed of the in-wheel motors and r_i is the effective rotation radius of tire.

The consumed power and regenerative braking power of in-wheel motors can be calculated by

$$P_{ei}=T_{ea}\omega_{xi}{\eta_e}^{-\mathrm{sgn}(T_{ea})} \tag{22}$$

where P_{ei} is the power of each motor; η_e is the efficiency of in-wheel motors. The motor efficiency map fitted according to the experimental results is shown in Fig. 3.

The hydraulic braking system of 6WID UGV is mainly composed of hydraulic cylinder, and each electric wheel is equipped with a brake cylinder. According to the experimental results, the braking torque generated by the braking system is approximately linear with the brake cylinder pressure. Considering the response of the hydraulic system, the braking force can be expressed as

$$T_{bi}=\frac{\sigma\bullet P_{iref}}{\tau_b s+1} \tag{23}$$

where $T_{bi}(i=fl,fr,ml,mr,rl,rr)$ is the hydraulic braking torque provided by the hydraulic braking system; τb is the time constant; σ is the gain; $P_{iref}(i=fl,fr,ml,mr,rl,rr)$ is the reference hydraulic pressure.

3. Path tracking and torque distribution controller

When the electro-hydraulic steering system fails, the differential-steering based path tracking control of 6WID UGV

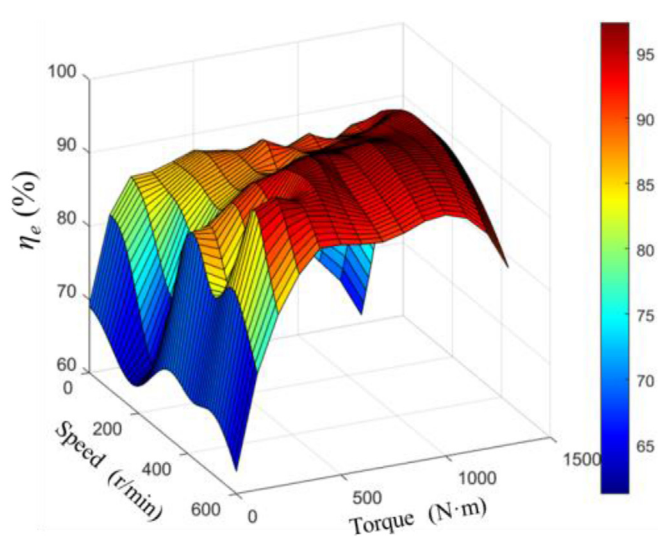


Fig. 3. Efficiency map of in-wheel motor.

needs to consider not only the tracking accuracy, but also the energy efficiency, especially the yaw stability when the in-wheel motor fault occurs. Therefore, this paper adopts hierarchical control scheme. The overall scheme of torque distribution strategy of 6WID UGV based on differential steering mode is shown in Fig. 4.

According to the desired model, the upper controller estimates the required yaw moment through model free adaptive sliding mode control (MFASMC). The required total longitudinal force is obtained by PI control. The lower controller reasonably distributes the generalized control force and yaw moment by using seeker optimization algorithm (SOA) to each in-wheel motor and hydraulic braking system of 6WID UGV to optimize the energy efficiency and ensure the accuracy of path tracking. Only the input and output data are required for path tracking of the 6WID UGV. The state parameters required in the control scheme can be obtained by state observer or sensors. The proposed method does not rely on accurate mathematical model and can be applied in practice.

3.1. The upper controller

Based on the accurate mathematical model, the yaw moment required by differential steering can be calculated. Unfortunately, according to **Eq. (15)**, 6WID UGV is a strongly nonlinear system. Because parameters such as cornering stiffness are time-varying, it is difficult to obtain an accurate mathematical model. Sliding mode control (SMC) is a control method with simple structure and superior control performance. It is widely used in energy saving [29,30], trajectory control [31,32] and fault tolerant control [33,34]. However, the design of the SMC controller still needs the actual parameters of the controlled system [35]. Because 6WID UGV is a strong nonlinear system, especially when performing tasks in the field, it is difficult to establish an accurate mathematical model, so the model-based control strategy may be difficult to obtain satisfactory performance. As a consequence, in this section, the model free adaptive control (MFAC) scheme is combined with SMC, and a model free adaptive sliding mode control (MFASMC) method is proposed to estimate the required yaw moment for the path tracking of 6WID UGV based on differential steering.

3.1.1. Theory of MFAC

Before using MFAC method, the following assumptions are made for single input single output (SISO) discrete-time nonlinear system.

Assumption 1. Except for finite time points, the partial derivatives of $f(\cdots)$ with respect to the (n_y+2)th variable is continuous.

Assumption 2. Nonlinear system satisfies the generalized Lipschitz condition, that is

$$|y(k_1+1)-y(k_2+1)|\leqslant b\|\mu(k_1)-\mu(k_2)\| \tag{24}$$

where nonlinear system $y(k_i+1) = f(y(k_i),\cdots,y(k_i-n_y),\mu(k_i),\cdots,\mu(k_i-n_u))$, $i=1{,}2$; b is a positive constant.

According to **Assumption 1** and **Assumption 2**, there are the following theorems [36].

Theorem 1. *If a SISO discrete-time nonlinear system satisfies assumptions 1 and 2, when $\|\Delta\mu(k)\|\neq 0$, there must exist a time-varying parameter $\varphi_c(k)\in R^m$ named the PPD that causes the system to transform into the compact-form dynamic linearization (CFDL) model of* **Eq. (25)**:

$$y(k+1)=y(k)+\varphi_c^T(k)\Delta\mu(k) \tag{25}$$

where $\varphi_c(k)$ is bounded at any time, and $\Delta\mu(k) = \mu(k)-\mu(k-1)$, $\Delta y(k+1)=y(k+1)-y(k)$.

Compact-form dynamic linearization model free adaptive control (CFDL-MFAC) of SISO discrete-time nonlinear systems is as follows

$$\widehat{\varphi}_c(k)=\widehat{\varphi}_c(k-1)+\frac{\eta\Delta\mu(k-1)(\Delta y(k)-\widehat{\varphi}_c(k-1)\Delta\mu(k-1))}{u+\|\Delta\mu(k-1)\|^2} \tag{26}$$

$$|\widehat{\varphi}_c(k)|<\varepsilon \text{ or } |\Delta\mu(k-1)|\leq\varepsilon \text{ or } sign(\widehat{\varphi}_c(k))\neq sign(\widehat{\varphi}_c(1)),\ \widehat{\varphi}_c(k)=\widehat{\varphi}_c(1) \tag{27}$$

$$\mu(k)=\mu(k-1)+\frac{\rho\widehat{\varphi}_c(k)(y^*(k+1)-y(k))}{\lambda+\|\widehat{\varphi}_c(k)\|^2} \tag{28}$$

where $\widehat{\varphi}_c(k)$ is the estimation of the PPD $\varphi_c(k)$; $\widehat{\varphi}_t(1)$ is the initial value of the PPD, $i=1,\cdots,m$; $\lambda>0$ and $\mu>0$ are weight coefficients to adjust the change rates of the PPD and controller output, respectively; $\eta\in(0,2]$ and $\rho\in(0,1]$ are the step factors.

3.1.2. Calculation of required yaw moment based on MFASMC

Although MFAC method can control the system without relying on accurate mathematical model, it also has the disadvantage of slow response speed. Therefore, by combining MFAC and SMC , the required yaw moment is estimated. The sliding surface is defined as follows :

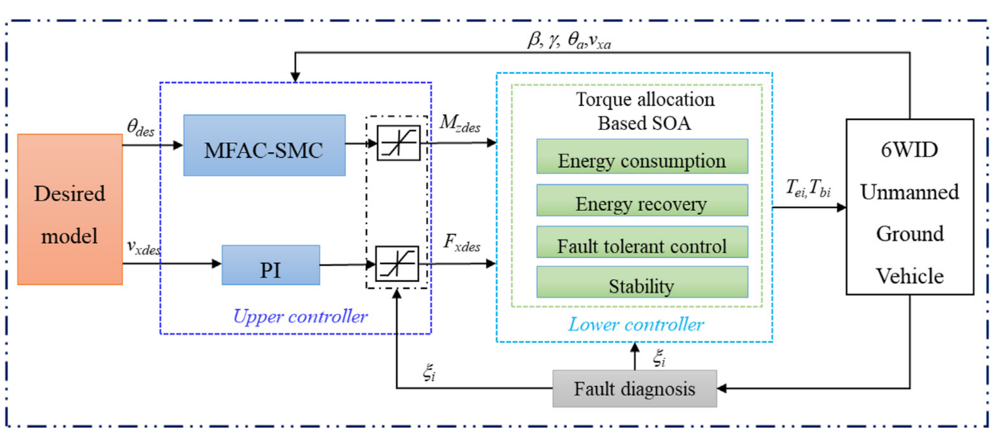


Fig. 4. The architecture of path tracking and torque distribution strategy based on differential steering.

$$\begin{aligned} S(k+1) &= c_1 e(k+1) + c_2 e(k) \\ &= c_1(\theta_{des}(k+1) - \theta(k+1)) + c_1(\theta_{des}(k) - \theta(k)) \end{aligned} \tag{29}$$

where c_1 and c_2 are positive coefficients.

The common approach rate of SMC can be expressed as

$$\dot{S} = -\varepsilon sat\left(\frac{S}{\psi}\right) - \alpha S \varepsilon > 0, \alpha > 0 \tag{30}$$

Eq. (31) is obtained by discretizing Eq. (30).

$$\frac{S(k+1) - S(k)}{T} = -\varepsilon sat\left(\frac{S(k)}{\psi}\right) - \alpha S(k) \tag{31}$$

where T is the sampling time; sat (x) is a saturation function; ψ is a small positive constant; α and ε are positive coefficients.

According to Eq. (29) and Eq. (31), Eq. (32) can be obtained.

$$\begin{aligned} &c_1(\theta_{des}(k+1) - \theta(k+1)) + c_2(\theta_{des}(k) - \theta(k)) \\ &\quad = -\varepsilon T sat\left(\frac{S(k)}{\psi}\right) - (\alpha T - 1) S(k) \end{aligned} \tag{32}$$

Let $\theta(k+1) = \theta(k) + \varphi_c(k)(M_{zdes}(k+1) - M_{zdes}(k))$, combine with Eq. (28), and add forgetting coefficient $0 < \kappa \le 1$ to the control law to improve the convergence characteristic of the algorithm, the yaw moment control law is obtained.

$$M_{zdes}(k+1) = \kappa M_{zdes}(k) + \frac{1}{c_1 \varphi_c(k)}\left[c_1 \theta_{des}(k+1) + c_2 \theta_{des}(k) - (c_1 + c_2)\theta(k) + \varepsilon T sat\left(\frac{S(k)}{\psi}\right) + (\alpha T - 1) S(k)\right] \tag{33}$$

Path tracking shall be carried out on the premise of ensuring the yaw stability of the vehicle. The yaw stability control often takes the yaw rate as the control target. The expected yaw rate with active steering mechanism in the steering process is a function related to the wheel angle [27]. However, the steering angle does not exist in the path tracking control based on differential steering. Therefore, the expected yaw moment is constrained to ensure the vehicle yaw stability. Meanwhile, the expected path obtained by the path planning algorithm ensures that there are no obstacles on the road and prevents vehicles from colliding or crashing.

Based on Eq. (15), the state equations of yaw rate and sideslip angle can be obtained

$$\begin{cases} \dot{\beta} = a_1 \beta + a_2 \gamma \\ \dot{\gamma} = a_3 \beta + a_4 \gamma + \dfrac{1}{I_z} M_z \end{cases} \tag{34}$$

Let $\dot{\beta} = 0$ and $\dot{\gamma} = 0$, the steady-state values of sideslip angle and yaw rate are respectively.

$$\beta_s = \frac{a_2}{I_z(a_1 a_4 - a_2 a_3)} M_z \tag{35}$$

$$\gamma_s = \frac{a_1}{I_z(a_2 a_3 - a_1 a_4)} M_z \tag{36}$$

Since the lateral force is provided by the tire, if the tire sideslip angle is too large, the vehicle may be unstable. Nevertheless, if there are too many constraints and the sideslip angle is very small or zero, it will increase the energy consumption. Accordingly, the desired yaw moment is constrained within the reasonable range of sideslip angle and yaw rate to guarantee stability.

$$M_{z\min} \le M_z \le M_{z\max} \tag{37}$$

$$M_{z\min} = \max\left(\left|\frac{I_z(a_1 a_4 - a_2 a_3)}{a_2}\right| \beta_{\min}, \left|\frac{I_z(a_2 a_3 - a_1 a_4)}{a_1}\right| \gamma_{\min}\right) \tag{38}$$

$$M_{z\max} = \min\left(\left|\frac{I_z(a_1 a_4 - a_2 a_3)}{a_2}\right| \beta_{\max}, \left|\frac{I_z(a_2 a_3 - a_1 a_4)}{a_1}\right| \gamma_{\max}\right) \tag{39}$$

$$|\gamma| \le \left|\sum u_i F_{zi}\right| \Big/ m v_x \tag{40}$$

where $M_{z\min}$ and $M_{z\max}$ represent the minimum and maximum values of yaw moment constraints respectively; $\beta_{\min}$ and $\beta_{\max}$ respectively represent the minimum and maximum values that can be reached when ensuring the stable operation of 6WID UGV; $\gamma_{\min}$ and $\gamma_{\max}$ represent the minimum and maximum values under the ground adhesion limit; u_i represents the friction coefficient of each wheel.

The required longitudinal force can be calculated by PI controller according to the desired speed of the upper controller

$$F_{xdes} = m a_{des} = m\left(k_p\left(v_{xdes} - v_x\right) + k_i \int\left(v_{xdes} - v_x\right) dt\right) \tag{41}$$

where F_{xdes} is the desired generalized longitudinal force; a_{des} is the expected acceleration, and $a_{des} \le ug$; v_{xdes} is the desired longitudinal speed; k_p and k_i are the parameters of PI controller.

When the motor fault occurs, the traction capacity of the whole vehicle should be taken account, so the constraint of the required longitudinal force is

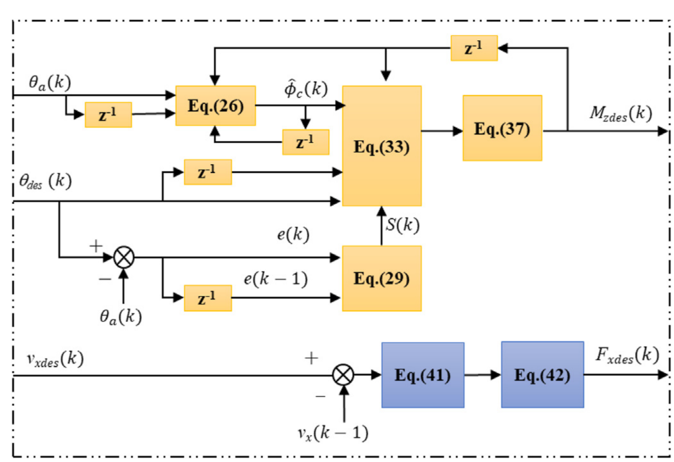


Fig. 5. The architecture of upper controller for path tracking.

$$\sum \xi_i \frac{T_{emin}}{R_{ref}} - \sum \xi_i T_b \le F_{xdes} \le \sum \xi_i \frac{T_{emax}}{R_{ref}} \tag{42}$$

where $0 \le \xi_i(i = fl, fr, ml, mr, rl, rr) \le 1$ is the failure factor of the in-wheel motor , which can be obtained from the fault diagnosis module, and the relevant research results can be referred to Refs. [37,38]; R_{ref} is the effective radius of the tire.

It is worth noting that the fault conditions considered in this paper are that the remaining motors that operate normally can still provide the desired longitudinal force and yaw moment, or the operational performance is relatively weakened within an acceptable range. Finally, the schematic diagram of the upper controller is shown in Fig. 5.

3.2. The lower controller

The yaw moment and generalized longitudinal force calculated by the upper controller is distributed to six in-wheel motors in the lower controller. The scheme of torque distribution strategy is illustrated in Fig. 6.

As shown in Fig. 6, the Seeker Optimization Algorithm (SOA) is used to distribute the optimal torque of six electric wheels. SOA learns from the experience of human society and simulates the intelligent behavior of human search activities to optimize the problem [39–41]. SOA has several actions, such as egoism action, altruistic action, proactive action and uncertain action. As the name suggests, egoism action is that as an independent individual in the intelligent group, each seeker believes in his own social experience and tends to move to his best historical position through cognitive learning. Altruistic action means that agents can share group social experience and constantly adjust their behavior to achieve common goals. Proactive action refers to the past action and results of the agent, which can predict and guide the future action. Uncertain action means that seekers can simulate human intelligent search action and have certain fuzzy logic. When the seeker is in a better position, it should search in a smaller area; when the location is poor, it should search in a larger area. Based on the above ideas, SOA can determine the search step and direction of agent, and optimize the problem. The specific torque distribution process is as follows.

3.2.1. Optimization objectives

The output of the control allocation can be expressed as

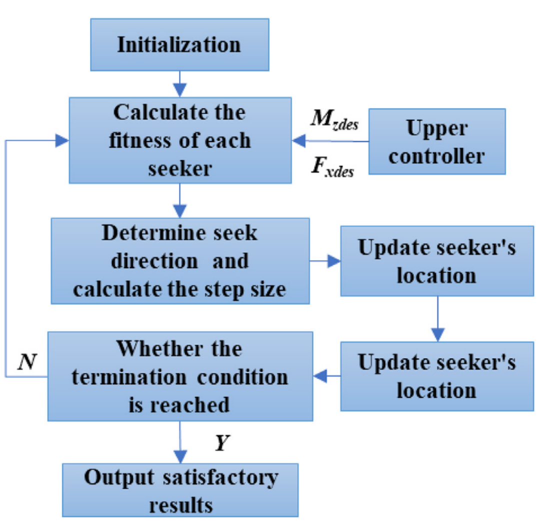


Fig. 6. The scheme of torque distribution strategy.

$$u_e = \left[T_{efl}, T_{efr}, T_{eml}, T_{emr}, T_{erl}, T_{err}\right]^T \tag{43}$$

$$u_{Tb} = \left[T_{bfl}, T_{bfr}, T_{bml}, T_{bmr}, T_{brl}, T_{brr}\right]^T \tag{44}$$

where u_e represents the vector composed of the torque provided by each wheel; u_{Tb} is a vector composed of the braking force provided by each hydraulic cylinder.

The stability and energy consumption in path tracking based on differential steering are very important for safety and range of 6WID UGV. In order to reduce energy consumption as much as possible on the basis of ensuring path tracking accuracy and stability, the following optimization functions are considered in this paper.

In order to ensure the tracking accuracy, the first optimization objective is

$$J_1 = \left| F_{xdes} - \frac{1}{R_{ref}} \sum_{i=fl}^{rr} \xi_i (u_{ei} - u_{Tbi}) \right|^2 + |M_{zdes} - b(F_{xr} - F_{xl})|^2 \tag{45}$$

The 6WID UGV requires sufficient energy to maintain high mobility when performing tasks in the field, and the second optimization goal is the power of energy saving and energy recovery.

$$J_2 = \sum P_i \tag{46}$$

In Eq. (46), if $T_{ei}>0$, $P_i > 0$; when $T_{ei}<0$, $P_i < 0$.

In order to ensure that one side of the vehicle has the same power characteristics as much as possible to prolong the working life of the motor, the third optimization objective can be expressed as

$$J_3 = \left(P_{fl} - P_{ml}\right)^2 + \left(P_{fl} - P_{rl}\right)^2 + \left(P_{fr} - P_{mr}\right)^2 + \left(P_{fr} - P_{rr}\right)^2 \tag{47}$$

Considering the real-time performance of the global optimization algorithm in the application, in order to reduce the search space, the following rules are considered in the torque distribution process: (1) on the premise of ensuring the tracking accuracy, improve the braking force as much as possible, and then maximize the distribution of regenerative braking force; (2) the insufficient braking force is provided by the hydraulic braking system; (3) meanwhile, considering the possible execution faults of the in-wheel motors during torque distribution, the priority should be given to stability rather than energy consumption. To sum up, the optimization objective function is obtained.

$$J = \begin{cases} W_1 J_1 + W_2 J_2 + W_3 J_3, \xi_i = 1 \\ J_1, 0 \le \xi_i < 1 \end{cases} \tag{48}$$

where W_1, W_2 and W_3 are weight coefficients.

3.2.2. Control allocation

*Step*1: Initialization. The maximum number of iterations is set to g. Generate N initial positions within the maximum torque range that can be provided by the motor and the hydraulic braking system.

$$u_{Te}(:,:,m) = \xi_{im} \left(\frac{T_{emax}}{R_{ref}} - \left(\frac{|T_{emin}|}{R_{ref}} + |F_{bmax}| \right) * rand() \right), 0 < m < N \tag{49}$$

*Step*2: Calculate the fitness of each position.

$$fitness(m)=\begin{cases}W_1J_1(\ u_{Te}(\ :,:,m))+W_2J_2(u_{Te}(\ :,:,m)\ \ )+W_3J_3(\ u_{Te}(\ :,:,m)),\xi_{im}=1\\ J_1(\ u_{Te}(\ :,:,m)),0\le\xi_{im}<1\end{cases} \tag{50}$$

*Step*3 : Determination of the search strategy and calculate the search direction d_{mj} and step α_{mj} of each individual in the j-th iterative search process.

Through the analysis of egoism action, altruistic action and proactive action, the egoism research direction $d_{m,ego}$, altruism research direction $d_{m,alt}$ and proactive research direction $d_{m,pro}$ of m-th individuals are obtained.

$$d_{m,ego}(g)=p_{i,best}-u_{Te,j}(\ :,:,m) \tag{51}$$

$$d_{m,alt}(g)=g_{i,best}-u_{Te,j}(\ :,:,m) \tag{52}$$

$$d_{m,pro}(g)=u_{Te,j_1}(\ :,:,m)-u_{Te,j_2}(\ :,:,m) \tag{53}$$

where $g_{m,best}$ is the best position in collective history; $p_{m,best}$ is the best position experienced by the m-th search individual; $u_{Te,j_1}(:,:,m)$ and $u_{Te,j_2}(:,:,m)$ are the best positions of{ $u_{Te,j-2}(:,:,m)$, $u_{Te,j-1}(:,:,m)$, $u_{Te,j}(:,:,m)$} respectively.

The seeker comprehensively considers various factors and uses the randomly weighted geometric average of three directions to determine the search direction.

$$d_{mj}=sign\left(\omega d_{m,pro}+\varphi_1 d_{m,ego}+\varphi_2 d_{m,alt}\right) \tag{54}$$

where sign $(\cdot)$ is a symbolic function; $\varphi_1$and φ_2 are real numbers randomly selected in the range of [0,1]; ω is the weight coefficient, which decreases linearly from 0.9 to 0.1 with the increase of the number of iterations, and $\omega=(g\text{-}j)/g$.

Through the approximation ability of fuzzy system, the relationship between the fitness and step length can be established, and then the step length can be determined. This paper studies the minimization problem. The fuzzy rules of search action are described as follows: the smaller the fitness, the smaller the search step. Gaussian membership function is used to represent fuzzy variable of the step length.

$$u_f(x)=\exp\left[-(x-u)^2\Big/2\delta^2\right] \tag{55}$$

where u_f(x) represents the degree of membership function of the input variable x; u and δ are parameters of membership function.

The linear membership function is used to make the membership degree proportional to the order of the fitness. The best position has the maximum membership degree $u_{f\max}=1$, and the worst position has the minimum membership degree $u_{f\min}=0.0111$, assuming that the membership degree $u_f<0.0111$ can be ignored. The seeker in other positions has the membership degree $u_f<1$. The membership degree u_{mj} of the fitness of the individual m in the j-th iteration can be expressed as:

$$u_{mj}=rand(u_m,1),0<j<g \tag{56}$$

$$u_m=u_{\max}-\frac{N-I_m}{N-1}(u_{\max}-u_{\min}) \tag{57}$$

where u_m is the membership degree of the individual m; I_m is the sequence number of the corresponding fitness value of $u_{Te}(:,:,m)$ which are arranged in descending order.

Eq. (56) is used to simulate the randomness of human search behavior, and it is in the range of [u_m,1]. According to Eq. (57), the step length can be obtained as:

$$\alpha_{mj}=\delta_{mj}\sqrt{-\ln(u_{mj})} \tag{58}$$

where α_{mj} is the step length of individual m in the j-th iteration;δ_{mj} is the parameter of Gaussian membership function, and it can be expressed as following.

$$\delta_{mj}=\omega\bullet\left|u_{Te,j}(\ :,:,\max)-u_{Te,j}(\ :,:,\min)\right| \tag{59}$$

where $u_{Te,j}$(:,:,max) and $u_{Te,j}$ (:,:,min) are the positions with minimum and maximum fitness in the same group respectively.

*Step*4 : Position update. The update equation is as following.

$$\Delta u_{Te,j}(\ :,:,m)=\alpha_{mj}d_{mj} \tag{60}$$

$$u_{Te,j+1}(\ :,:,m)=u_{Te,j}(\ :,:,m)+\Delta u_{Te,j}(\ :,:,m) \tag{61}$$

*Step*5 : If the number of iterations reaches the maximum or meets the termination conditions, the optimization stops and the historical optimal position is output. Otherwise, return to the step 2.

4. Simulation results

In this section, the control model and desired model are established in MATLAB/Simulink for co-simulation with TruckSim. The feasibility and effectiveness of the proposed path tracking method and torque distribution strategy are verified by simulation under two driving conditions of *S*-curve and straight line. In order to show that the proposed control scheme has good energy-saving characteristics, tracking performance and fault-tolerant control performance, the torque distribution strategy based on particle swarm optimization (PSO) [42–45] and the rule-based torque

Table 1
Main technical parameters of 6WID UGV in differential steering mode.

Parameters	Value
Mass (kg)	2500
Vertical distance from middle axle to front axle (m)	1.1–2.1
Distance from front axle to center of gravity (m)	1.6
Distance from rear axle to center of gravity (m)	1.6
Effective radius of tire (m)	0.46
Maximum power of motor (kW)	30
Maximum output torque of motor within 30s (N·m)	2400
Maximum off-road speed (km/h)	35
Maximum speed (km/h)	60

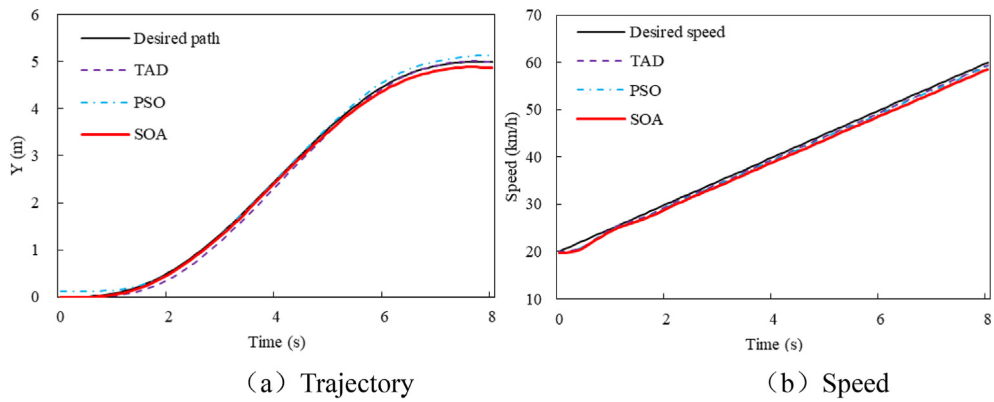


Fig. 7. Trajectory and speed.

Table 2
RMS errors of path tracking.

Road condition	Allocation strategy		
	TAD	PSO	SOA
0.4	0.10 m	0.09 m	0.08 m
0.8	0.09 m	0.08 m	0.08 m

average distribution (TAD) control strategy are selected for comparative analysis. The main technical parameters of 6WID UGV in differential steering mode are shown in Table 1.

4.1. The performance of path tracking and energy saving

In order to analyze the influence of different torque distribution strategies on the upper path tracking control, three cases of acceleration, constant speed and deceleration are selected for simulation verification. By calculating the energy consumption and energy recovery rate under different torque distribution strategies, the energy-saving characteristics of different distribution strategies can be compared. During the simulation, the distance from the middle axle to the front axle is set to 1.6 m, and the road friction coefficient is set to 0.8.

Case1. acceleration mode

Fig. 7 shows the trajectory and speed under different torque

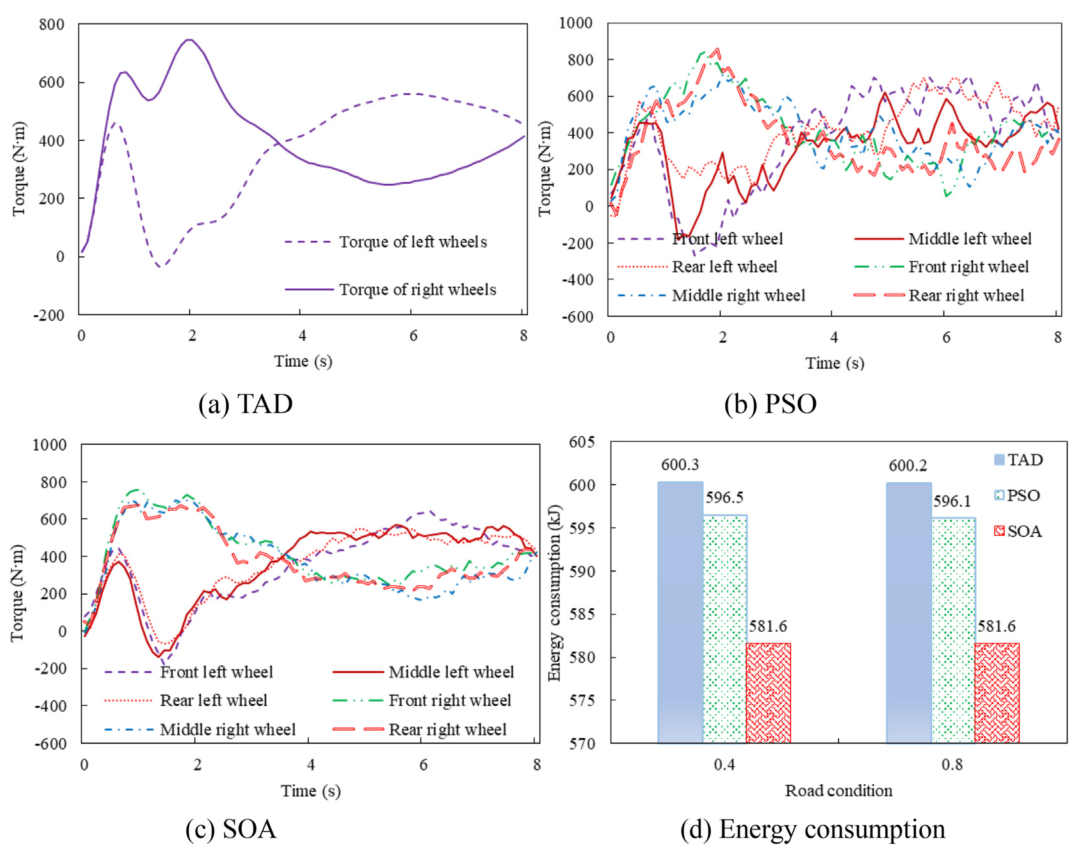


Fig. 8. Torque of different strategies and energy consumption.

Table 3
Energy consumption under different road conditions.

Road condition	Allocation strategy		
	TAD	PSO	SOA
0.4	0 (base)	0.63%	3.11%
0.8	0 (base)	0.68%	3.09%

Table 4
RMS errors of path tracking.

Road condition	Allocation strategy		
	TAD	PSO	SOA
0.4	0.15 m	0.44 m	0.15 m
0.8	0.14 m	0.15 m	0.10 m

distribution strategies. It can be seen from Fig. 7 that the three distribution methods can better meet the requirements of the MFASMC controller and realize the path tracking. The root mean square (RMS) error of path tracking is shown in Table 2. The different distribution schemes have small tracking error and can adapt to different road adhesion coefficients, indicating the effectiveness of torque distribution strategy and the robustness of MFASMC.

The output torque of different distribution methods and the energy consumption under different road conditions are illustrated in Fig. 8. Fig. 8 (a) shows the output torque under TAD strategy. The characteristics of motors on both sides of 6WID UGV are consistent, which means the TAD strategy is simple and easy to implement. Fig. 8 (b) is the torque optimized by PSO method. PSO adopts the idea of global optimization, and the characteristics of each motor are quite different. Fig. 8 (c) indicated the output torque based on the SOA strategy adopted in this paper. The torque of motors on both sides has the same change trend as that of TAD method. In this paper, energy consumption is considered as the optimization object, and in the process of differential steering, the motors on both sides of the vehicle will have different speed, so that the torque of the motors on the same side is also different. Ignoring the energy loss between other mechanical structures, the energy consumption under different road adhesion is shown in Table 3. Compared with PSO and TAD methods, SOA allocation strategy has better energy-saving characteristics when meeting the requirements of the upper controller. Therefore, the torque distribution method based on SOA has better control performance and energy-saving characteristics.

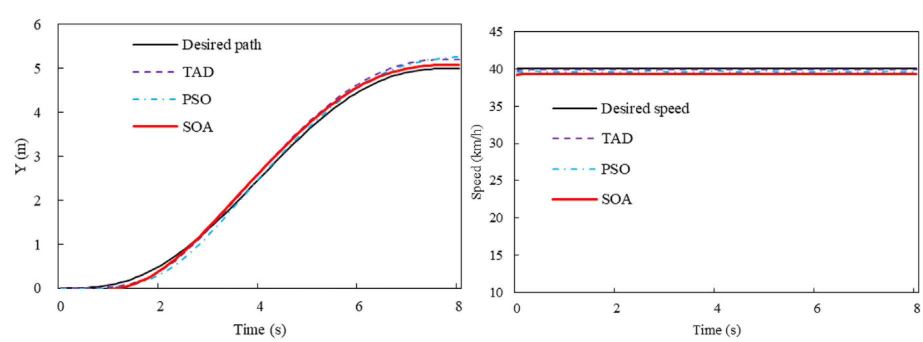
Case2. constant speed mode

The simulation results are carried out under the condition of constant speed operation, and the trajectory and speed shown in Fig. 9 are obtained. The RMS error of path tracking is shown in Table 4. It can be seen from Table 4 that the path tracking accuracy under constant speed condition is lower than that acceleration mode. The main reason is that longitudinal force required under uniform speed condition is small, and the allocation strategy requires better performance. The control performance of PSO method for low adhesion is reduced. Since the longitudinal force required is small, the PSO method is easy to fall into local optimization and control difficulty is increased. SOA method and TAD method can better adapt to the changes of road adhesion.

The torque and energy consumption are shown in Fig. 10. Without considering other energy losses, the energy consumption under different road adhesion is shown in Table 5. It can be seen from Fig. 9 and Table 5: (1) TAD and SOA strategy make the motor have relatively better output characteristics; (2) when the road adhesion coefficient decreases, PSO has better energy-saving characteristics, but the path tracking performance decreases; (3) the torque distribution method based on SOA can not only realize the coordinated control of traction and differential steering, but also has better path tracking performance and better energy-saving characteristics.

Case3. deceleration mode

In order to verify the coordinated control performance of the different torque distribution strategies for braking and differential steering, the deceleration condition is selected for simulation test, and the results are shown in Fig. 11 and Fig. 12. The RMS errors of path tracking and energy recovery characteristics are shown in



(a) Trajectory (b) Speed

Fig. 9. Trajectory and speed.

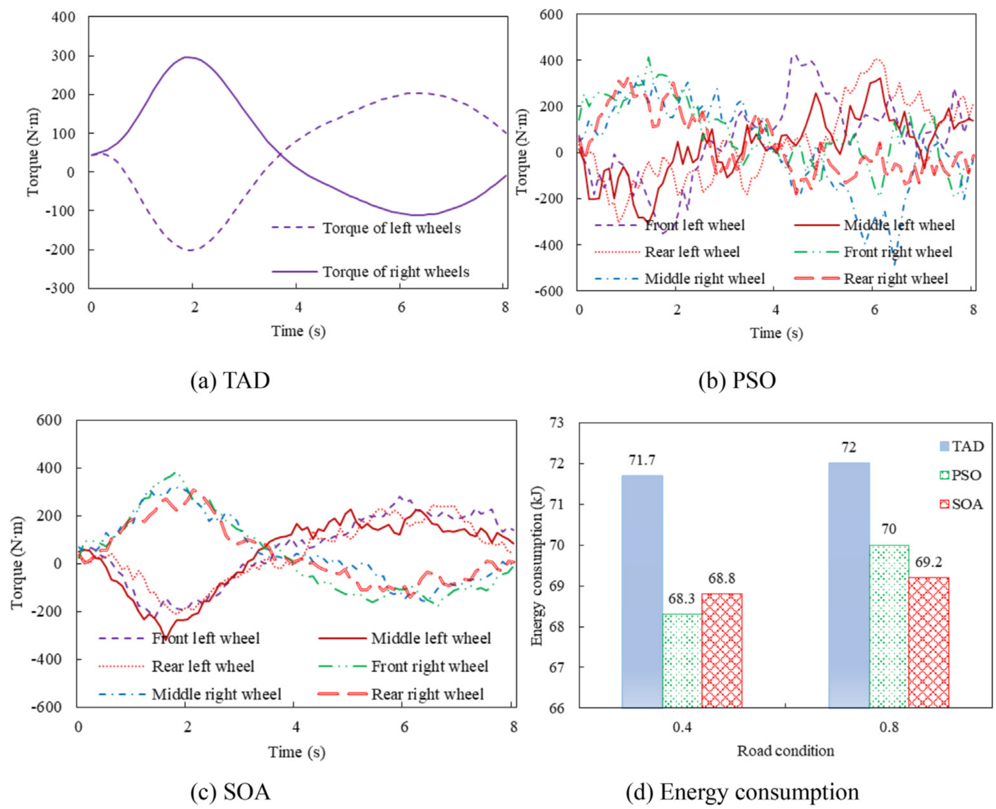


Fig. 10. Torque of different strategies and energy consumption.

Table 5
Energy consumption under different road conditions.

Road condition	Allocation strategy		
	TAD	PSO	SOA
0.4	0 (base)	4.74%	4.04%
0.8	0 (base)	2.85%	3.89%

Table 6 and Table 7. The following conclusions can be drawn from these simulation results: (1) the three control methods can realize the coordinated control of braking and differential steering, and adapt to the change of road adhesion coefficient; (2) compared with TAD and PSO strategy, SOA strategy can improve braking energy recovery and achieve the purpose of energy saving.

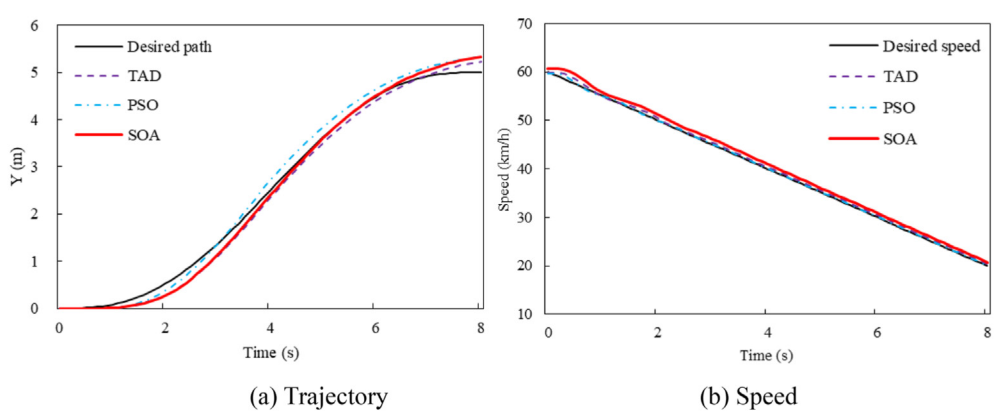


Fig. 11. Trajectory and speed.

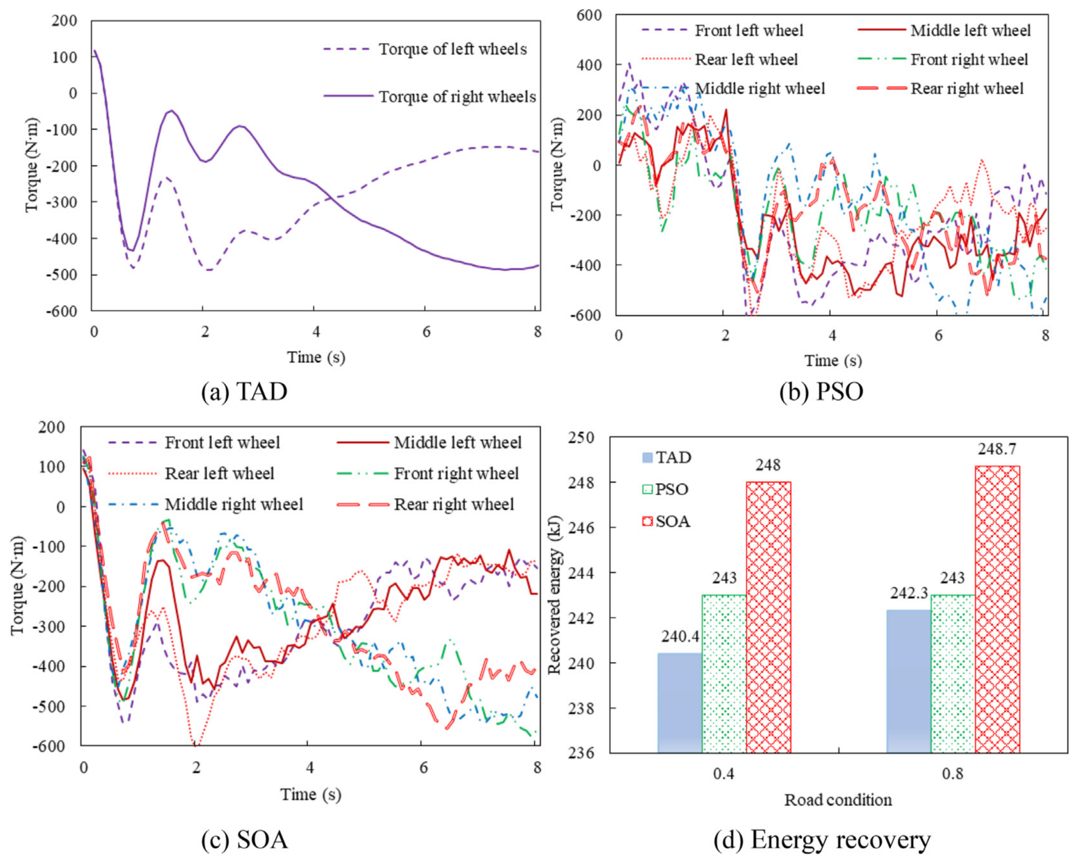


Fig. 12. Torque and energy recovery.

Table 6
RMS errors of path tracking.

Road condition	Allocation strategy		
	TAD	PSO	SOA
0.4	0.14 m	0.17 m	0.18 m
0.8	0.16 m	0.17 m	0.16 m

Table 7
Energy recovery under different road conditions.

Road condition	Allocation strategy		
	TAD	PSO	SOA
0.4	0 (base)	1.08%	3.16%
0.8	0 (base)	0.29%	2.64%

4.2. *The performance of fault-tolerant control*

In order to further verify the feasibility and effectiveness of SOA strategy for fault-tolerant control performance, the simulation results are carried out assuming that the right rear wheel of the 6WID UGV fails completely during operation. The vehicle speed shall be maintained at a constant speed of 40 km/h. The friction coefficient is 0.4.

(1) Straight line

The fault-tolerant control results under straight-line operation are shown in Fig. 13. Fig. 13 (a) shows the lateral offset of the vehicle. The output torque of the right rear wheel motor is zero at 5s. The path deviation without fault-tolerant control gradually increases, while the SOA method can still maintain a small tracking error. Fig. 13 (b) shows the yaw rate. The yaw rate of SOA strategy is less than that without fault-tolerant control. Fig. 13 (c) shows the motor torque using TAD distribution strategy in case of motor fault, but fault-tolerant control is not considered. Therefore, when the lateral error increases, the upper controller attempts to generate a corrected yaw moment, but the lower torque distribution does not consider fault-tolerant control, so the lateral error increases all the time, which also shows the effectiveness of the upper controller. Fig. 13 (d) illustrates the effectiveness of SOA method with fault-tolerant control. When the motor fails, it can still track the desired path and ensure the driving stability of the vehicle under the condition of meeting the traction requirements.

The fault-tolerant control performance of the *S*-curve path

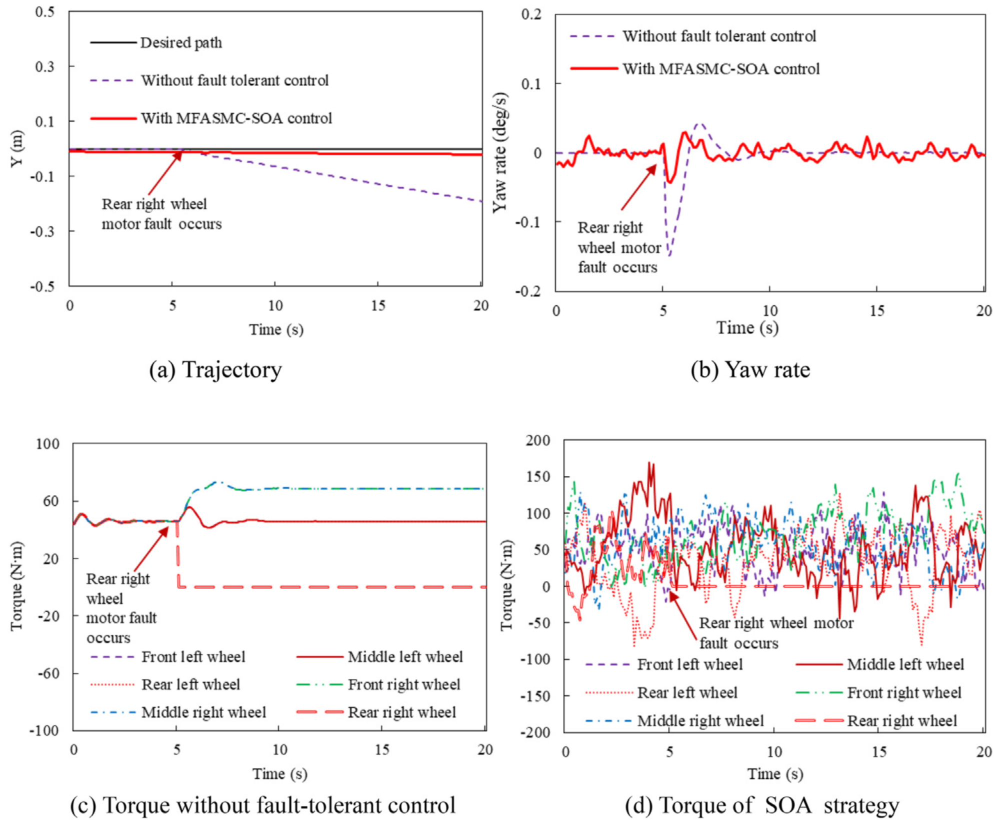


Fig. 13. Fault tolerant control of straight-line operation.

(2) *S*-curve

tracking is shown in Fig. 14. According to Fig. 14 (a) and Fig. 14 (b), the SOA strategy can effectively track the desired path and ensure the stability of the vehicle regardless of motor fault. Fig.14 (c) and Fig. 14 (d) illustrate that the SOA method with fault tolerance control can reasonably distribute torque and ensure yaw stability.

The strategy proposed in this paper only needs input and output data to design the controller, which simplifies the complexity of the controller and gains better control performance. Therefore, it also provides a useful reference for the vehicles that are difficult to obtain accurate mathematical, systems with the simple single-track scheme or vehicles that moves in safe and limited speed, such as amphibious vehicles, tracked vehicles, the piece-holder trolleys used in the industrial sector. Meanwhile, the strategy of this paper also has some limitations, such as the requirement of accurate motor efficiency map and advanced fault diagnosis module, but with the development of advanced fault detection technology, these problems can also be overcome.

5. Conclusion

This paper studied the path tracking and torque distribution strategy of 6WID UGV in differential steering mode, and optimizes the energy consumption and energy recovery rate on the basis of ensuring vehicle stability. The following conclusions can be obtained through simulation results.

(1) The proposed MFASMC-SOA strategy is feasible and effective for path tracking of 6WID UGV based on differential steering;
(2) The torque distribution strategy with the fault-tolerant control can effectively ensure the vehicle stability;
(3) Compared with the allocation strategy of TAD and PSO, the allocation strategy based on SOA not only has better energy efficiency, but also ensures the path tracking accuracy, which provides a new solution to improve the energy efficiency and lays a foundation for the lightweight design of the 6WID UGV.

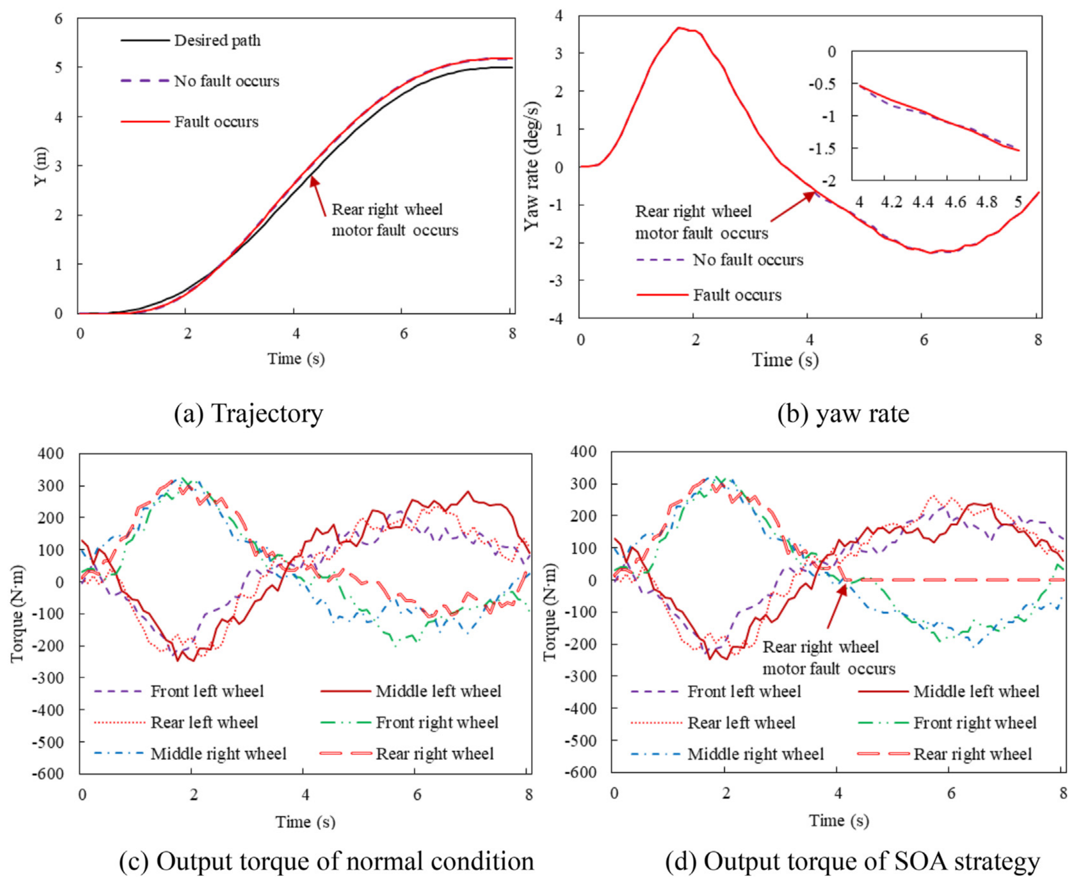


Fig. 14. Fault tolerant control of *S*-curve operation.

In the future work, we will continue to consider optimizing the efficiency of engine and generator, and the coordinated control method of independent steering, driving and braking under high-speed conditions.

Credit author statement

Jiang Yue: Conceptualization, Software, Writing – original draft. **Meng Hao:** Writing- Reviewing and Editing **Chen Guanpeng:** Visualization, Writing- Reviewing and Editing. **Yang Congnan:** Software. **Xu Xiaojun:** Project administration. **Zhang Lei:** Project administration, Visualization, Investigation. **Xu Haijun:** Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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